

ME 2261 - Embedded Systems Log Book**1D Array**

```
1  #define F_CPU 8000000UL
2  #include <avr/io.h>
3  #include <util/delay.h>
4
5  int main(void)
6  {
7      DDRD = 0xFF; PORTD = 0x00;
8      int n[8]; n[0] = 1;
9
10     for (int i=1; i<8; i++)
11     {
12         n[i] = n[i-1]*2;
13         PORTD = n[i-1] + n[i];
14     }
15 }
```

Functions

```
1  #define F_CPU 8000000UL
2  #include <avr/io.h>
3  #include <util/delay.h>
4
5  int factorial(int n);
6
7  int main(void)
8  {
9      int n = 5;
10     int v;
11     v = factorial(n);
12     return v;
13
14 }
15
16 int factorial(int n){
17     int product = 1;
18     for (int i = 1; i<=n; i++)
19         product = product * i;
20     return product;
21 }
```

Task 1

Write program that displays the factorial of a number on an LCD, based on input from the serial monitor.

Write program that displays the factorial of a number on an PORT C, based on input from 8 bits connected to PORT D. Consider the number input in binary form.

```
17 int factorial(int n);
18
19 int main(void)
20 {
21     DDRD = 0x00;
22     DDRC = 0xFF;
23     int n = PORTD;
24     int v;
25     v = factorial(n);
26     PORTC = v;
27     return 1;
28 }
29
30 int factorial(int n){
31     int product = 1;
32     for (int i = 1; i<=n; i++)
33         product = product * i;
34     return product;
35 }
```

Name	Address	Value	Bits
I/O PIND	0x30	0x00	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐
I/O DDRD	0x31	0x00	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐
I/O PORTD	0x32	0x05	☐ ☐ ☐ ☐ ☒ ☒ ☒ ☒

Name	Address	Value	Bits
I/O PINC	0x33	0x78	☐ ☒ ☒ ☒ ☒ ☐ ☐ ☐
I/O DDRC	0x34	0xFF	☒ ☒ ☒ ☒ ☒ ☒ ☒ ☒
I/O PORTC	0x35	0x78	☐ ☒ ☒ ☒ ☒ ☐ ☐ ☐

For input of PORTD = 5, output of PORTC = 120 was obtained. [5! = 120]

Pointer

main.c

Run

Output

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int *pc, c;
6     c = 22;
7     pc = &c;
8     printf("Address of variable c :%u", &c);
9     printf("\nValue of variable c :%u", c);
10    printf("\nAddress of pointer pc :%u", &pc);
11    printf("\nValue of pointer pc :%u", pc);
12    printf("\nValue pointed by pointer pc :%u", *pc);
13
14    return 0;
15 }
```

Address of variable c :156133764
Value of variable c :22
Address of pointer pc :156133768
Value of pointer pc :156133764
Value pointed by pointer pc :22

=== Code Execution Successful ===

Example: -

```
int main(void)
{
    DDRB = 0xFF; PORTB = 0x00;
    int numbers[] = {10, 20, 30, 40, 50};
    int *ptr = numbers;

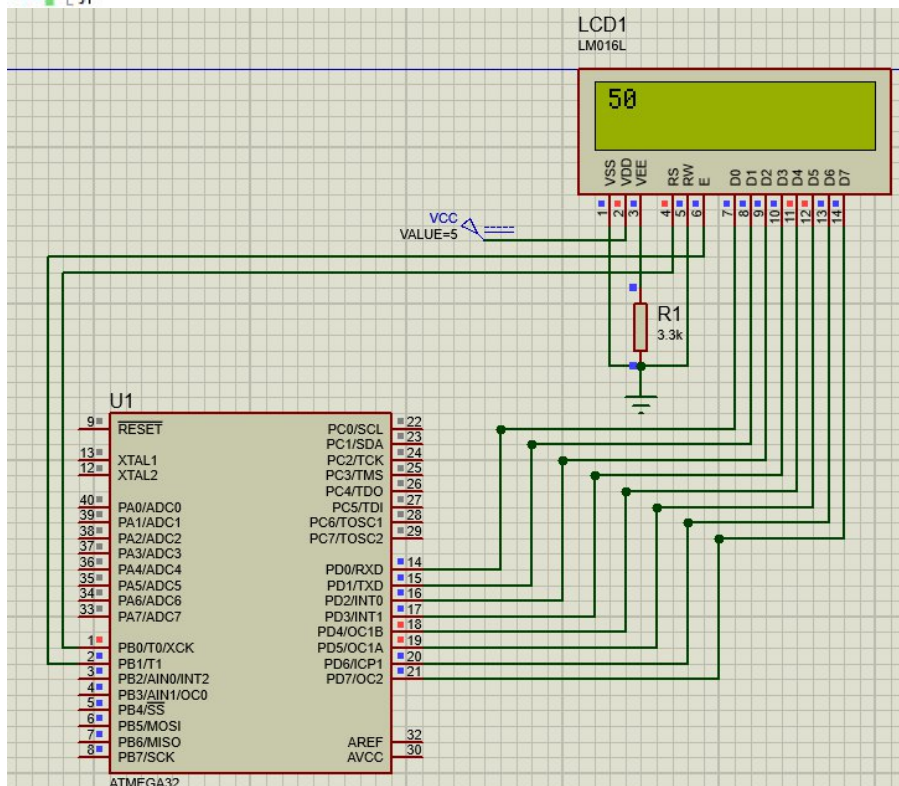
    for (int i=0; i<5; i++)
    {
        PORTB = *ptr;
        ptr++;
    }
}
```

Name	Address	Value	Bits
I/O PINB	0x36	0x32	
I/O DDRB	0x37	0xFF	
I/O PORTB	0x38	0x32	

Task 2

Count number from 1 to 50 with 500 ms delay in the LCD.

```
1  #define F_CPU 8000000UL
2  #include <avr/io.h>
3  #include <stdlib.h>
4  #include <util/delay.h>
5  #define D0 eS_PORTD0
6  #define D1 eS_PORTD1
7  #define D2 eS_PORTD2
8  #define D3 eS_PORTD3
9  #define D4 eS_PORTD4
10 #define D5 eS_PORTD5
11 #define D6 eS_PORTD6
12 #define D7 eS_PORTD7
13 #define RS eS_PORTB0
14 #define EN eS_PORTB1
15 #include "lcd.h"
16
17 int main(void)
18 {
19     DDRD = 0xFF;
20     DDRB = 0xFF;
21     Lcd8_Init();
22     char Stri[4];
23     for (int i = 1; i <= 50; i++)
24     {
25         itoa(i, Stri, 10);
26         Lcd8_Set_Cursor(1,0);
27         Lcd8_Write_String(Stri);
28         _delay_ms(500);
29         Lcd8_Clear();
30     }
31 }
```



Task 3

Write a program that displays the following results.

Display 'UOM' and shift it to the left and then to the right.

Count down from 10 to 0 (with 1s delay)

Finally, display 'Start LAB' and make it blink two times.

```
17 int main(void)
18 {
19     DDRD = 0xFF;
20     DDRB = 0xFF;
21     Lcd8_Init();
22     char Stri[4];
23     while(1){
24         Lcd8_Set_Cursor(1,7);
25         Lcd8_Write_String("UOM");
26         for (int j=0;j<7;j++)
27         { Lcd8_Shift_Left();
28           _delay_ms(250);}
29         for (int j=0;j<13;j++)
30         { Lcd8_Shift_Right();
31           _delay_ms(250);}
32         for (int j=0;j<5;j++)
33         { Lcd8_Shift_Left();
34           _delay_ms(250);}
35         Lcd8_Clear();
36         for (int i = 10;i>=0; i--)
37         { itoa(i, Stri, 10);
38           Lcd8_Set_Cursor(1,0);
39           for (int j=0;j<7;j++)
40           {Lcd8_Shift_Right();}
41           Lcd8_Write_String(Stri);
42           _delay_ms(500);
43           Lcd8_Clear(); }
44         Lcd8_Set_Cursor(1,4);
45         Lcd8_Write_String("Start LAB");
46         _delay_ms(500);
47         Lcd8_Clear();
48         _delay_ms(500);
49         Lcd8_Set_Cursor(1,4);
50         Lcd8_Write_String("Start LAB");
51         _delay_ms(500);
52         Lcd8_Clear();
53         _delay_ms(500);
54         Lcd8_Set_Cursor(1,4);
55         Lcd8_Write_String("Start LAB");
56         _delay_ms(500);
57         Lcd8_Clear(); }
```

Task 1

Write program that displays the factorial of a number on an LCD, based on input from 8 switches connected to PORT A. Consider the number input in binary form using the switches.

```
int factorial(int n);
int main(void)
{
    DDRD = 0xFF;
    DDRC = 0xFF;
    DDRA = 0x00;
    Lcd8_Init();
    char Stri[4];
    while(1){
        int n = PINA;
        _delay_ms(500);
        int v = factorial(n);
        itoa(n,Stri,10);
        Lcd8_Set_Cursor(1,0);
        Lcd8_Write_String("Fibonacci of ");
        Lcd8_Write_String(Stri);
        Lcd8_Set_Cursor(2,6);
        itoa(v,Stri,10);
        Lcd8_Write_String(Stri);
        _delay_ms(1000);
        Lcd8_Clear();
        _delay_ms(500);
    }
}

int factorial(int n){
    int product = 1;
    for (int i = 1; i<=n; i++)
        product = product * i;
    return product;
}
```